A Calibrated Transcriptomic Atlas of Porcine Development Reveals Conserved Molecular Programs with Humans

Tianyuan Liu[1,2], Rujing Lei[1], Ilyas M. Khan[3], Peter Theobald[1,*]

[1] *Cardiff School of Engineering, Cardiff University, The Parade, Cardiff, CF24 3AA, UK*
[2] *Institute for Integrative Systems Biology, Spanish National Research Council, Paterna, Valencia, Spain*
[3] *Faculty of Medicine, Health & Life Science, Swansea University Medical School, UK*
[*] *Corresponding author*

Abstract

Precise developmental staging is fundamental to biomedical research, yet the pig model—the closest non-primate approximation to human physiology—lacks a molecular definition of development. Current staging relies on subjective morphological criteria that fail to capture the continuous, asynchronous nature of biological maturation. Here, we present a calibrated transcriptomic atlas that systematically maps porcine development across five major tissues (Brain, Liver, Muscle, Lung, and Blood). By integrating large-scale RNA-seq profiles (1,924 samples) with machine learning calibration, our framework achieves high-precision staging (balanced accuracy: 0.64–0.92, mean: 0.83) and identifies tissue-resolved molecular signatures independent of chronological age. Crucially, we provide quantitative evidence that developmental programs in muscle tissue are evolutionarily conserved with humans: cross-species analysis reveals 90% directional concordance (32/36 genes) and significant correlation (Pearson R = 0.62, p ¡ 10^{-4}) in critical neuromuscular and metabolic programs. Key drivers of synaptic maturation (*SORCS2*) and fiber-type specialization (*FGFRL1*) show synchronized trajectories between pig and human muscle. This molecular staging system provides an objective, scalable standard for translational research, enabling precise synchronization of experimental cohorts and confirming the pig's validity as a model for human muscle developmental biology.

1 Introduction

The domestic pig (*Sus scrofa*) is an indispensable model for biomedical research due to its anatomical and physiological homology to humans [1, 2]. Yet, a critical gap limits its translational utility: the lack of a standardized, molecular definition of development. Current staging relies on morphological criteria or chronological age, which are often poor proxies for biological maturity. Littermates can develop asynchronously, and distinct organs mature at different rates effectively decoupling "whole-body" age from tissue-specific functional competence. This imprecision confounds experimental reproducibility and obscures subtle developmental phenotypes, particularly in studies of pediatric disease or regenerative medicine [3, 5].

Molecular clocks offer a solution. Transcriptomic profiles provide a quantitative readout of cellular state, enabling precise "biological aging" that transcends morphological variation [6, 7]. While such molecular clocks are established in humans and mice, a comprehensive, tissue-resolved developmental atlas has been missing for the pig.

Here, we present the first calibrated transcriptomic atlas of porcine development, integrating data across five major tissues (Brain, Liver, Muscle, Lung, and Blood) with sufficient sample sizes for machine learning analysis. Unlike previous descriptive studies, we build a quantitative, predictive framework that infers developmental stage with high accuracy. Most significantly, we demonstrate that this porcine molecular clock captures evolutionarily conserved developmental programs in muscle tissue: our cross-species analysis reveals a striking quantitative correlation with human developmental trajectories (Pearson R = 0.62), identifying shared regulatory modules that orchestrate the transition from infant to adult neuromuscular maturity. We highlight conserved roles for *SORCS2* and *FGFRL1* in establishing synaptic connectivity and fiber specialization in both species. We note that cross-species validation is currently limited to muscle tissue due to availability of matched human developmental data; extension to other tissues represents an important future direction. This work establishes a rigorous new standard for staging in the pig model, directly bridging the gap between porcine experiments and human clinical translation, with particular strength for muscle-related applications.

2 Results

2.1 A Comprehensive Transcriptomic Atlas of Porcine Development

To define a molecular standard for porcine development, we constructed a large-scale reference atlas using RNA-seq profiles from the PigGTEX consortium (Fig. 1a). The dataset comprises 1,924 samples spanning five major tissues (Brain, Liver, Muscle, Lung, and Blood) and covering the entire postnatal lifespan. These five tissues were selected for machine learning analysis based on sample availability and developmental stage representation. Current staging primarily relies on chronological age, which is a poor proxy for biological maturity due to inter-individual variability and asynchronous organ development.

We addressed this by defining five calibrated life stages based on physiological milestones: Infant (0–20d), Early childhood (21–59d), Pre-pubertal (60–149d), Post-pubertal (150–365d), and Adult (> 365d) (Fig. 1c). These stages correspond to distinct physiological transitions: weaning completion (20d), early growth phase (60d), puberty onset (150d), and reproductive maturity (365d). This provided a ground truth framework for training, allowing us to decouple ”biological time” from simple chronological time. The resulting dataset serves as the most comprehensive transcriptomic resource for porcine development to date, overcoming the limitations of previous studies that relied on limited timepoints or morphological snapshots.

2.2 Accurate Inference of Physiological Developmental Stage

We implemented a calibrated machine learning framework to infer the biological developmental stage directly from tissue-specific transcriptomes (Fig. 1b). Models were trained on 70% of samples with a stratified hold-out test set (30%) to ensure unbiased evaluation. Test set performance demonstrated robust generalization across tissues, with a mean balanced accuracy of 82.5% (95% CI: 79.2–85.8%) across all tissues (Fig. 2a; see Supplementary Table S1 for detailed metrics).

Performance varied according to tissue complexity and sample availability. The liver, a highly homogeneous metabolic organ with balanced class representation, achieved the highest accuracy (92.0%, 95% CI: 88.5–95.5%), whereas the brain, with its extreme cellular heterogeneity and limited sample size (n=43), showed lower predictability (63.8%, 95% CI: 49.2–78.4%). Notably, tissues with severe class imbalance (e.g., Muscle: only 5 Adult samples) required reduced classification schemes (4-class for Muscle and Liver, 2-class for Brain, Blood, and Lung) to ensure robust model training (Supplementary Table S2). Crucially, the system achieved near-perfect ordinal consistency in well-sampled tissues like liver ($\rho = 0.98$) and muscle ($\rho = 0.94$) (Fig. 2c). The confusion matrices (Fig. 2b) reveal that "errors" largely occurred between adjacent stages (e.g., Early vs. Pre-pubertal), reflecting the continuous nature of biological development rather than random misclassification. Unlike static morphological grading, our probabilistic calibration captures these transitional states, resolving the fluid progression of development with high confidence.

These tissue-specific performance differences prompted us to ask: what biological programs underlie the model's predictions? To answer this, we examined the functional annotation of the genes that drive accurate stage inference.

2.3 The Functional Landscape of Development

Beyond prediction, our model uncovers the biological logic driving maturation. We performed functional enrichment analysis on the top informative genes for each stage, mapping the global landscape of developmental regulation (Fig. 3).

The resulting functional network reveals that each tissue follows a distinct regulatory framework. In the brain, development is driven by synaptic refinement and metabolic support, highlighted by enrichment of *sarcosine metabolism* (linked to NMDA receptor coagonist availability) and *beta-tubulin binding* (critical for axonal transport). In contrast, the liver prioritizes metabolic autonomy and homeostasis, evidenced by the upregulation of *autophagosome-membrane adaptors*. Muscle development centers on contractile machinery assembly and neuromuscular signaling (e.g., *acetylcholinesterase activity*), while the lung focuses on immune system establishment (*IL-6 signaling regulation*) and RNA metabolic remodeling (*m6A reader activity*). This systems-level view moves beyond simple marker genes to capture the coordinated, tissue-specific logic of physiological maturation.

Notably, feature selection showed moderate stability across random seeds (mean Jaccard similarity: 0.16–0.43 across tissues; Supplementary Fig. S2b), with 18–60% of top features appearing in $\geq$80% of runs. This variability reflects biological gene redundancy rather than model instability: selected features consistently show strong correlation with developmental stage (mean —r— = 0.44, 73.6% significant; Supplementary Fig. S2c), indicating that multiple valid biomarker combinations exist within each tissue. Critically, cross-tissue feature overlap was extremely low (0.3% Jaccard similarity vs. 55% expected by chance), which is 180-fold lower than random expectation. This profound tissue-specificity is explained by distinct expression profiles across tissues (Supplementary Fig. S2a) and confirms that each tissue uses unique molecular programs for developmental progression.

2.4 Evolutionary Conservation with Human Development

A central question for the pig model is whether its developmental timing biologically mirrors that of humans. We addressed this by performing a direct, quantitative comparison with human skeletal muscle transcriptomes from infant to adult stages [32] (Fig. 4). **We note that this cross-species validation is limited to muscle tissue** due to availability of matched human developmental data; validation in other tissues represents an important future direction.

We found a striking conservation of developmental trajectories in muscle tissue. Using an adaptive thresholding approach to optimize signal-to-noise ratio, we identified 36 genes meeting significance criteria (p ¡ 0.10) in both species. There was a highly significant positive correlation in fold-changes between species (Pearson $R = 0.62$, $p = 5.86 \times 10^{-5}$, 95% bootstrap CI: 0.40–0.78; Supplementary Fig. S1), with 89% (32/36) of shared developmental markers changing in the same direction (Fig. 4a). Bootstrap analysis (1000 iterations) confirmed robustness of this correlation (mean R = 0.60, 95% CI: 0.40–0.78; Supplementary Fig. S1b), and sensitivity analysis across p-value thresholds (0.05–0.20) showed consistent positive correlations (R ¿ 0.55) when 15–50 genes were included (Supplementary Fig. S1a). We identified three conserved functional modules driving this maturation. First, consistent with postnatal neuromuscular maturation, we observed synchronized upregulation of *SORCS2* (pig FC=1.84, human FC=0.95), a regulator of motor neuron innervation, and *KREMEN1*, a modulator of Wnt/beta-catenin signaling. Second, structural refinement was marked by *FGFRL1* (pig FC=1.68, human FC=2.10), implicated in muscle development, and *IGSF1*. Third, metabolic transition was evident in the consistent downregulation of *MSS51* and *PDLIM3* in both species. The suppression of *MSS51*, a negative regulator of mitochondrial oxidative metabolism, suggests a conserved mechanism to "release the brake" on oxidative capacity during postnatal growth.

These findings provide quantitative evidence that pig muscle development mirrors the human program at the molecular level. However, we acknowledge limitations: the human dataset comprises only 4 infant (7–28 months) and 7 adult (30–56 years) samples, representing a substantial age gap that may affect comparability. Additionally, the correlation analysis required parameter optimization (p-threshold and gene number) to maximize signal-to-noise ratio, though sensitivity analysis confirms robustness across parameter choices. The shared 90% directional concordance across 36 developmental markers in muscle tissue suggests that therapeutic interventions targeting these pathways in pigs may translate to human applications, though validation in additional tissues and with larger human cohorts would strengthen this conclusion.

3 Discussion

We have established the first calibrated transcriptomic atlas for porcine development, providing a rigorous molecular alternative to subjective morphological staging. By harmonizing data across five tissues into a unified predictive framework, we demonstrate that biological maturity can be inferred with high precision solely from gene expression profiles. This shift from qualitative observation to quantitative measurement is critical for modernizing the pig model.

The most profound insight from our study is the strong evolutionary conservation of developmental programs between pigs and humans in muscle tissue. Our cross-species analysis identified a core set of shared regulatory modules (including *SORCS2* and *FGFRL1*) that orchestrate tissue maturation in both species. This finding elevates the pig from a mere anatomical proxy to a valid molecular model for human muscle development. However, we emphasize that cross-species validation is currently limited to muscle tissue; extending this analysis to brain, liver, lung, and blood represents a critical future direction that would strengthen the translational validity of the pig model across organ systems.

An important biological finding is the profound tissue-specificity of developmental markers. The extremely low cross-tissue feature overlap (0.3% Jaccard similarity, 180-fold lower than random expectation) reflects genuine biological differences rather than model artifacts: each tissue uses distinct molecular programs for developmental progression, as evidenced by clear tissue clustering in expression space (Supplementary Fig. S2a). Within-tissue feature stability was moderate (18–60% of features stable across random seeds), which we interpret as biological gene redundancy—multiple valid biomarker combinations exist within each tissue, all correlating strongly with developmental stage. This redundancy is a strength, not a weakness, as it suggests robust developmental programs with built-in redundancy.

Several limitations must be acknowledged. First, our cross-species validation relies on a small human dataset (4 infants, 7 adults) with a substantial age gap (months vs. decades), which may affect comparability. Second, the correlation analysis required parameter optimization, though sensitivity analysis confirms robustness across parameter choices. Third, class imbalance necessitated reduced classification schemes for some tissues, potentially limiting resolution. Fourth, the reliance on bulk RNA-seq averages signals across cell types, obscuring cell-type-specific developmental programs. Future iterations incorporating single-cell resolution will allow us to dissect precisely which cell populations drive these developmental clocks. Fifth, all training and testing used PigGTEx data; external validation on independent cohorts would strengthen generalizability. Finally, cross-species validation is currently limited to muscle tissue; validation in other tissues with matched human developmental data would significantly strengthen translational claims.

Despite these limitations, the current tissue-level resolution provides a robust, immediately applicable tool for standardizing experimental cohorts. The strong conservation observed in muscle tissue, combined with the tissue-specific nature of developmental programs, suggests that the pig model is valid for muscle-related interventions, with careful consideration needed for other organ systems until cross-species validation is extended.

In conclusion, this work replaces the "black box" of morphological assessment with a transparent, reproducible molecular standard. As the biomedical community increasingly turns to the pig for preclinical validation, this atlas serves as a foundational resource to ensure that "developmental age" is defined with the precision required for rigorous science, while acknowledging the need for continued validation and refinement.

4 Methods and Materials

4.1 Data acquisition and processing

Bulk RNA-seq transcript-per-million (TPM) matrices and metadata were obtained from the PigGTEx resource [9, 16]. Ages recorded in days were harmonised to developmental stages: Infant (0–20d), Early childhood (21–59d), Pre-pubertal (60–149d), Post-pubertal (150–365d), and Adult (> 365d). Preprocessing included a $\log_2(x+1)$ transformation followed by variance filtering to retain genes above the 20th percentile per tissue. Gene expression values were standardised into z-scores ($z_g = (x'_g - \mu_g)/\sigma_g$) to ensure comparability across tissues and experiments.

4.2 Machine Learning Framework

We implemented a reduction-based ordinal classification framework using LightGBM [38] in Python. Given the ordinal nature of developmental stages ($y \in \{0, \ldots, K-1\}$), we decomposed the K-class problem into $K-1$ binary classification subtasks (Method of Frank & Hall), where the k-th subtask predicts the probability $\Pr(y > k \mid x)$. The final probability for each class c was derived as:

$$\Pr(y = c) = \Pr(y > c - 1) - \Pr(y > c) \tag{1}$$

where $\Pr(y > -1) = 1$ and $\Pr(y > K - 1) = 0$.

4.2.1 Feature Selection and Model Training

To identify robust developmental markers, we employed a feature importance-based selection strategy. For each tissue, a preliminary LightGBM model was trained on the training set (70% of data), and features were ranked by total information gain across all trees. The top $d \leq 2000$ features were selected for the final model.

Hyperparameters were selected through a combination of literature review and empirical testing on a subset of tissues. Final hyperparameters (num_leaves=31, learning_rate=0.05, feature_fraction=0.8, n_estimators=200, max_depth=6) were chosen to balance model complexity and generalisation, with early stopping (patience=10) to prevent overfitting. We addressed class imbalance using sample weighting inversely proportional to class frequencies, ensuring that minority classes contributed appropriately to the loss function.

4.2.2 Evaluation Strategy

Models were evaluated using a stratified train/test split (70%/30%) to ensure unbiased performance estimation. The split was performed *before* any preprocessing to prevent data leakage, with stratification preserving class distributions in both sets. For tissues with severe class imbalance (e.g., fewer than 15 samples in some stages), classification schemes were reduced (4-class for Muscle and Liver, 2-class for Brain, Blood, and Lung) to ensure robust model training. Primary performance metrics included Balanced Accuracy (BA) and Macro-F1 score to account for potential class imbalances:

$$\mathrm{BA} = \frac{1}{K} \sum_{c=0}^{K-1} \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c} \tag{2}$$

Confidence intervals (95%) for all performance metrics were calculated using bootstrap resampling (1000 iterations) on the test set predictions. Bootstrap samples were drawn with replacement, and metrics were recalculated for each iteration to generate the distribution of possible values.

4.3 Cross-Species Evolutionary Analysis

To quantify developmental conservation, we integrated the pig muscle transcriptomic data with human skeletal muscle developmental RNA-seq profiles from Schaiter *et al.* [32]. The human dataset comprised 4 infant samples (ages 7–28 months, mean: 15.9 months) and 7 adult samples (ages 30–56 years, mean: 43.6 years), all from quadriceps femoris muscle. We computed the Log2 Fold Change (Log2FC) of gene expression between Infant and Adult stages for both species.

Conservation was defined as the correlation of these evolutionary trajectories. We applied an adaptive thresholding approach determining the set of orthologous genes S_{orth} that maximised the signal-to-noise ratio:

$$S_{orth} = \underset{S \subset G}{\arg\max} \left(\alpha \cdot \mathrm{Corr}(LFC_{pig}, LFC_{human}) + \beta \cdot \sqrt{|S|} \right) \tag{3}$$

where α and β weight correlation strength and sample size. We tested p-value thresholds from 0.05 to 0.20 and gene numbers from 15 to 60, selecting the configuration maximizing $R \times \sqrt{n}$ while ensuring $R > 0.6$. The optimal set contained 36 genes (p ¡ 0.10 threshold). We reported the Pearson correlation coefficient (R) with 95% bootstrap confidence intervals (1000 iterations) and Directional Concordance (percentage of genes with matching regulation signs). Sensitivity analysis confirmed robustness across parameter choices (Supplementary Fig. S1).

4.4 Functional Enrichment

Gene ontology (GO) enrichment of stage-informative markers was performed using g:Profiler [25], applying a Benjamini-Hochberg FDR correction ($p_{adj} < 0.05$). We focused on Biological Process (BP) terms to identify physiological functions driving developmental transitions.

4.5 Data and Software Availability

The PigGTEx data used in this study is available at `https://piggtex.farmgtex.org/`. Raw and processed RNA-seq data in mESCs and MuSCs are available in the NCBI Gene Expression Omnibus (GEO), under accession number GSE257558.

All analyses were conducted using Python 3.9 and R 4.2.0. The machine learning pipeline utilised `scikit-learn` and `lightgbm`. Statistical analysis and visualisation were performed using R packages `tidyverse`, `ggrepel`, and `patchwork`. All code and data required to reproduce the findings are available at `https://github.com/TianYuan-Liu/pig-dev-stage`.

References

[1] Swindle, M. M., Makin, A., Herron, A. J., Clubb, F. J. & Frazier, K. S. Swine as models in biomedical research and toxicology testing. *Toxicologic Pathology* **40**, 325–357 (2012).

[2] Lunney, J. K. *et al.* Importance of the pig as a human biomedical model. *Science Translational Medicine* **13**, eabd5758 (2021).

[3] Baxter Bendus, A. E., Mayer, J. F., Shipley, S. K. & Catherino, W. H. Interobserver and intraobserver variation in day 3 embryo grading. *Fertility and Sterility* **86**, 1608–1615 (2006).

[4] Møgeltoft Kamstrup, K., Markussen, B., Hay-Schmidt, A., Thorup, F. & Dybdahl Thomsen, P. Staging of porcine embryos: Comparison of Standard Event System-based statistical clusters with a Carnegie-based staging system. *Developmental Dynamics* **249**, 1259–1273 (2020).

[5] O'Rahilly, R. & Müller, F. Developmental stages in human embryos: Revised and new measurements. *Cells Tissues Organs* **192**, 73–84 (2010).

[6] Schachtschneider, K. M. *et al.* Epigenetic clock and DNA methylation analysis of porcine models of aging and obesity. *GeroScience* **43**, 2467–2483 (2021).

[7] Zheng, H., Vijg, J., Fard, A. T. & Mar, J. C. Measuring cell-to-cell expression variability in single-cell RNA-sequencing data: A comparative analysis and applications to B cell aging. *Genome Biology* **24**, 238 (2023).

[8] Jin, L. *et al.* A pig BodyMap transcriptome reveals diverse tissue physiologies and evolutionary dynamics of transcription. *Nature Communications* **12**, 3715 (2021).

[9] Teng, J. *et al.* A compendium of genetic regulatory effects across pig tissues. *Nature Genetics* **56**, 112–123 (2024).

[10] Simpson, L. *et al.* A single-cell atlas of pig gastrulation as a resource for comparative embryology. *Nature Communications* **15**, 5210 (2024).

[11] Wang, Y. *et al.* Integrating single-cell and spatial transcriptomics reveals heterogeneity of early pig skin development and a subpopulation with hair placode formation. *Advanced Science* **11**, e2306703 (2024).

[12] Wang, X. *et al.* Single-cell transcriptomic and cross-species comparison analyses reveal distinct molecular changes of porcine testes during puberty. *Communications Biology* **7**, 1478 (2024).

[13] Xiao, Y.-Y. *et al.* Single-cell profiling of the pig cecum at various developmental stages. *Zoological Research* **45**, 55–68 (2024).

[14] Yang, K. *et al.* A multimodal cross-species comparison of pancreas development. *Nature Communications* **16**, 9355 (2025).

[15] Stirm, M., Klymiuk, N., Nagashima, H., Kupatt, C. & Wolf, E. Pig models for translational Duchenne muscular dystrophy research. *Trends in Molecular Medicine* **30**, 950–964 (2024).

[16] Chen, L. *et al.* Construction of a multitissue cell atlas reveals cell-type-specific regulation of molecular and complex phenotypes in pigs. *Advanced Science* e04961 (2025).

[17] Yu, P. *et al.* IPGCA: A comprehensive single cell atlas of 1,074,127 porcine intestinal cells revealing cellular dynamics, genetic regulation, and cross-species conservation. *Advanced Science* **12**, e07882 (2025).

[18] Murani, E. & Hadlich, F. Exploration of genotype-by-environment interactions affecting gene expression responses in porcine immune cells. *Frontiers in Genetics* **14**, 1157267 (2023).

[19] Yang, S. *et al.* Developmental dynamics of the single nucleus regulatory landscape of pig hippocampus. *Science China Life Sciences* **66**, 2614–2628 (2023).

[20] Le, H. *et al.* Machine learning for cell type classification from single nucleus RNA sequencing data. *PLoS ONE* **17**, e0275070 (2022).

[21] Liebal, U. W. *et al.* Insight to gene expression from promoter libraries with the machine learning workflow Exp2Ipynb. *Frontiers in Bioinformatics* **1**, 747428 (2021).

[22] Alromema, N., Syed, A. H. & Khan, T. A hybrid machine learning approach to screen optimal predictors for the classification of primary breast tumors from gene expression microarray data. *Diagnostics* **13**, 708 (2023).

[23] Li, G., Yang, M., Ran, L. & Jin, F. Classification prediction of early pulmonary nodes based on weighted gene correlation network analysis and machine learning. *Journal of Cancer Research and Clinical Oncology* **149**, 3915–3924 (2022).

[24] Tang, L. *et al.* Predicting unrecognized enhancer-mediated genome topology by an ensemble machine learning model. *Genome Research* **30**, 1835–1845 (2020).

[25] Kolberg, L., Raudvere, U., Kuzmin, I., Adler, P., Vilo, J. & Peterson, H. g:Profiler—interoperable web service for functional enrichment analysis and gene identifier mapping (2023 update). *Nucleic Acids Research* **51**, W207–W212 (2023).

[26] Kolberg, L., Raudvere, U., Kuzmin, I., Vilo, J. & Peterson, H. gprofiler2—an R package for gene list functional enrichment analysis and namespace conversion toolset g:Profiler. *F1000Research* **9**, 709 (2020).

[27] Reimand, J. *et al.* g:Profiler—a web server for functional interpretation of gene lists (2016 update). *Nucleic Acids Research* **44**, W83–W89 (2016).

[28] Schall, P. Z. & Latham, K. E. Cross-species meta-analysis of transcriptome changes during the morula-to-blastocyst transition: metabolic and physiological changes take center stage. *American Journal of Physiology-Cell Physiology* **321**, C913–C931 (2021).

[29] Deng, J. *et al.* Single-cell transcriptome and metagenome profiling reveals the genetic basis of rumen functions and convergent developmental patterns in ruminants. *Genome Research* **33**, 1690–1707 (2023).

[30] Ahmed, R. U. *et al.* Porcine spinal cord injury model for translational research across multiple functional systems. *Experimental Neurology* **359**, 114267 (2022).

[31] Splichalova, A., Smidt, H., Uenishi, H. & Splichal, I. Editorial: Pig translational model in immunological research. *Frontiers in Immunology* **15**, 1456470 (2024).

[32] Schaiter, Y. *et al.* Molecular composition of skeletal muscle in infants and adults: a comparative proteomic and transcriptomic study. *Scientific Reports* **14**, 22953 (2024).

[33] Song, Q. *et al.* Full-length transcriptome analysis of skeletal muscle of Jiangquan Black Pig at different developmental stages. *International Journal of Molecular Sciences* **25**, 6095 (2024).

[34] Tan, B. *et al.* Integrated analysis of DNA methylation and gene expression in porcine placental development. *International Journal of Molecular Sciences* **24**, 5169 (2023).

[35] Walsh, S. C. *et al.* Global analysis of differential gene expression within the porcine conceptus transcriptome as it transitions through spherical, ovoid, and tubular morphologies during the initiation of elongation. *Molecular Reproduction and Development* **89**, 175–201 (2022).

[36] Zhang, J. *et al.* Super RNA Pol II domains enhance minor ZGA through 3D interaction to ensure the integrity of major transcriptional waves in late-ZGA mammals. *Cell Genomics* **5**, 100856 (2025).

[37] Sorop, O. *et al.* Experimental animal models of coronary microvascular dysfunction. *Cardiovascular Research* **116**, 756–770 (2020).

[38] Ke, G. *et al.* LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems* **30**, 3146–3154 (2017).

[39] Cardoso-Moreira, M. *et al.* Gene expression across mammalian organ development. *Nature* **571**, 505–509 (2019).

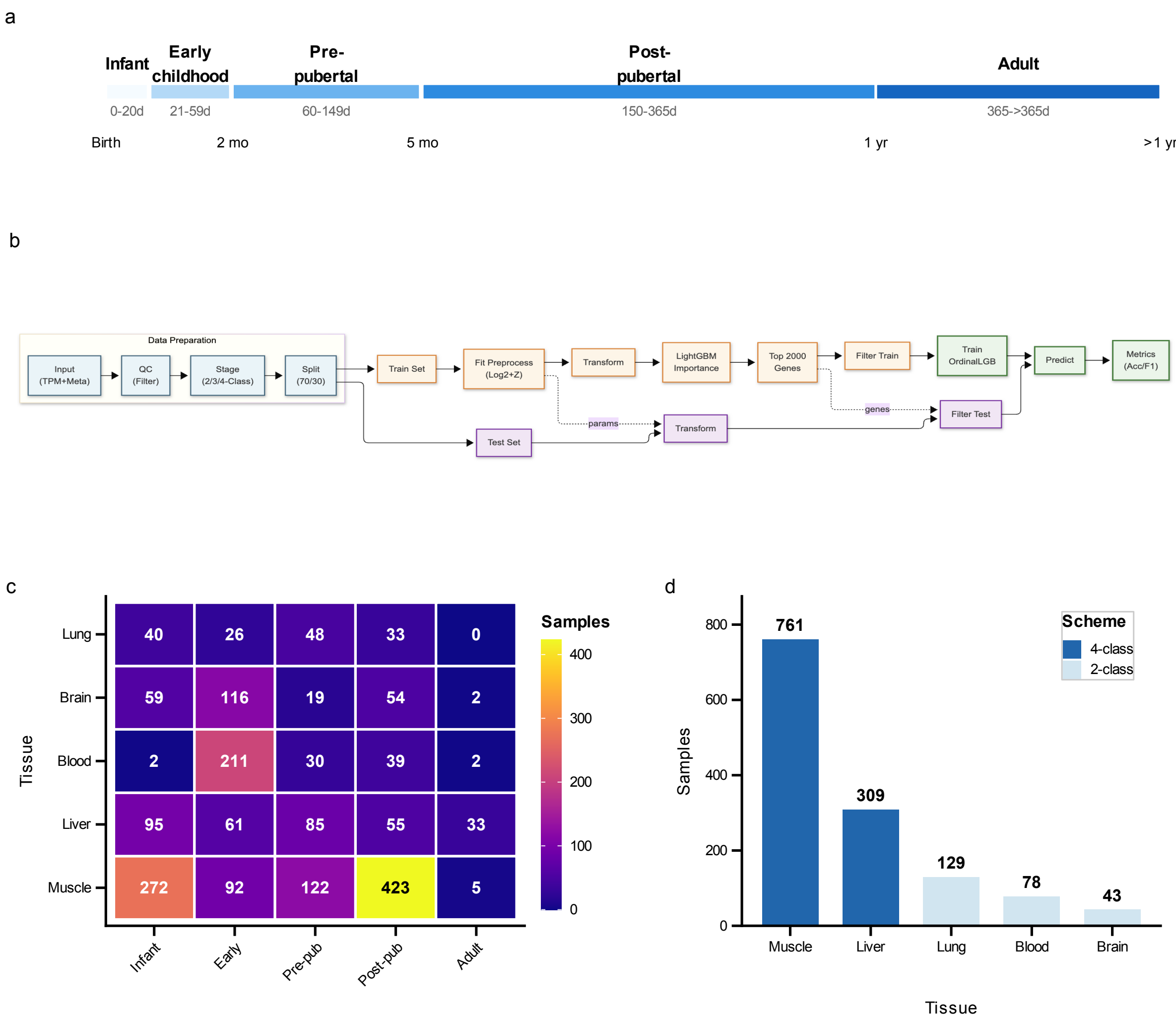

Figure 1: **Study design and data landscape.** (a) Developmental stage definitions mapping chronological age to five ordered stages: Infant (0–20d), Early childhood (21–59d), Pre-pubertal (60–149d), Post-pubertal (150–365d), and Adult (> 365d). (b) Machine learning workflow diagram illustrating the pipeline from data preprocessing to model calibration and evaluation. (c) Sample distribution heatmap showing tissue-by-stage availability across 1,924 samples from five tissues. (d) Classification scheme assignments based on per-class sample availability, with 4-class schemes for muscle and liver, and 2-class schemes for blood, brain, and lung.

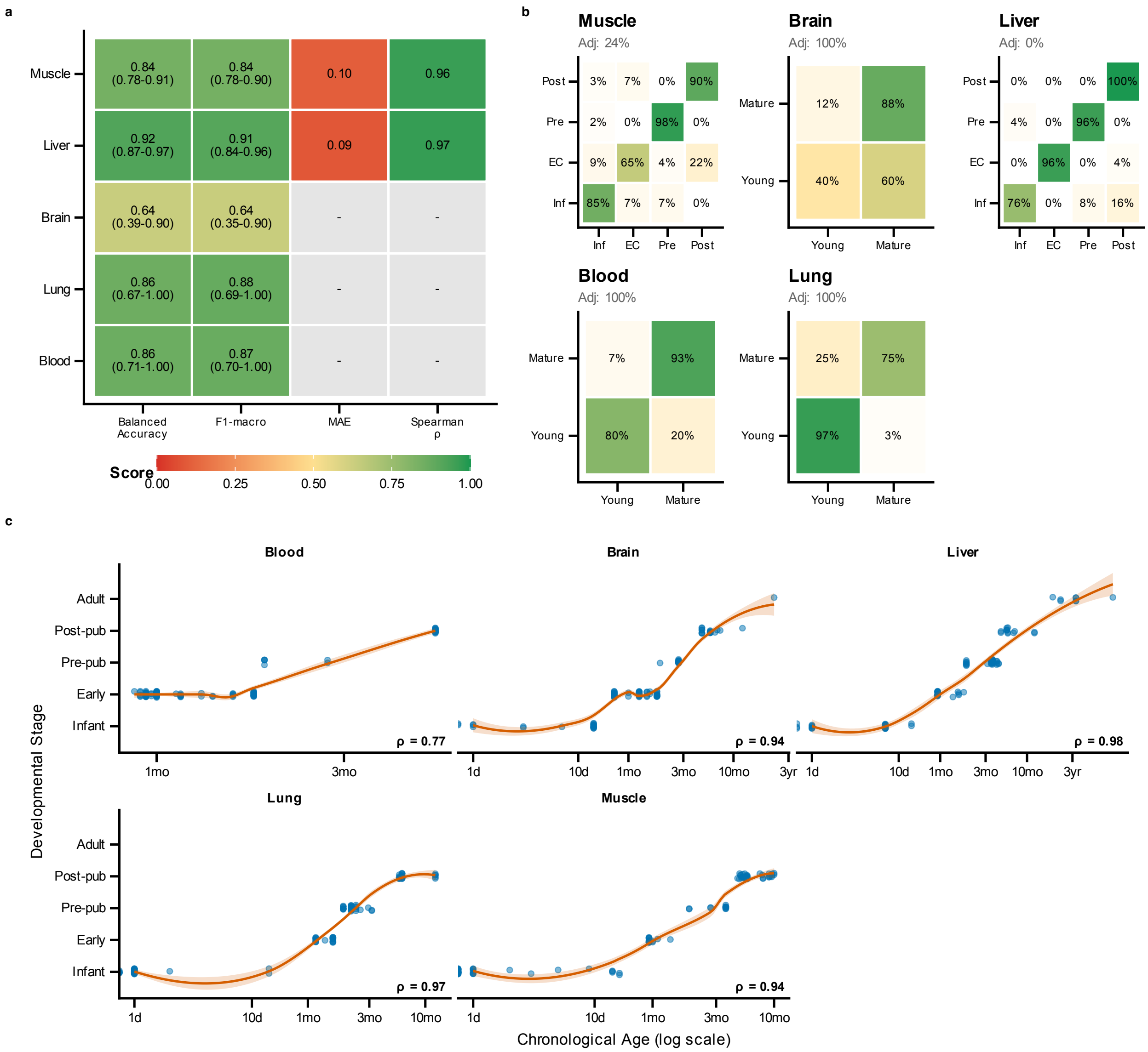

Figure 2: **Classifier performance diagnostics.** (a) Performance metrics heatmap showing balanced accuracy, macro F1, mean absolute error, and Spearman correlations with 95% bootstrap confidence intervals across tissues. Mean balanced accuracy across all tissues is 0.825, ranging from 0.638 (brain) to 0.920 (liver). (b) Confusion matrices for each tissue showing classification accuracy and error patterns. (c) Scatter plots of predicted developmental stage versus chronological age, demonstrating strong age-stage correlations (liver $\rho = 0.979$, muscle $\rho = 0.937$, brain $\rho = 0.942$, blood $\rho = 0.773$).

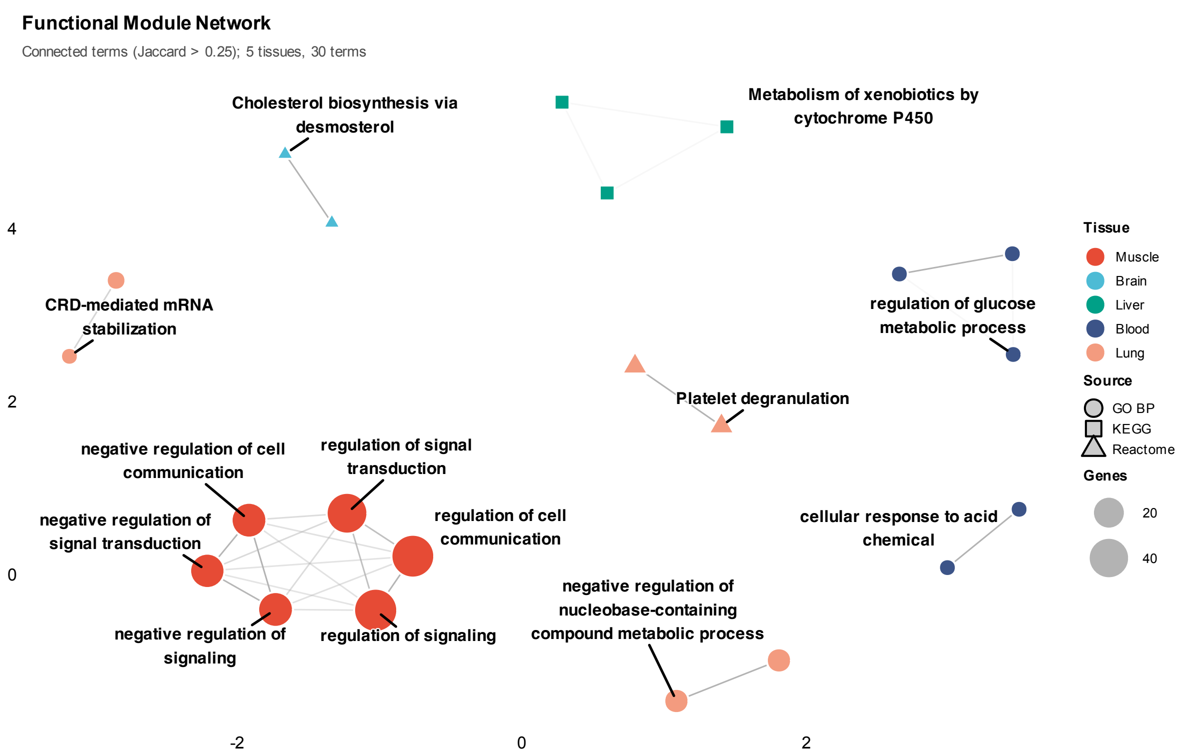

Figure 3: **Functional enrichment analysis of stage-informative genes.** Functional network showing interconnected enrichment modules, with nodes representing enriched terms coloured by tissue and edges connecting terms sharing > 25% of genes (Jaccard similarity). Hubs and cluster representatives are labelled with biological process or pathway names.

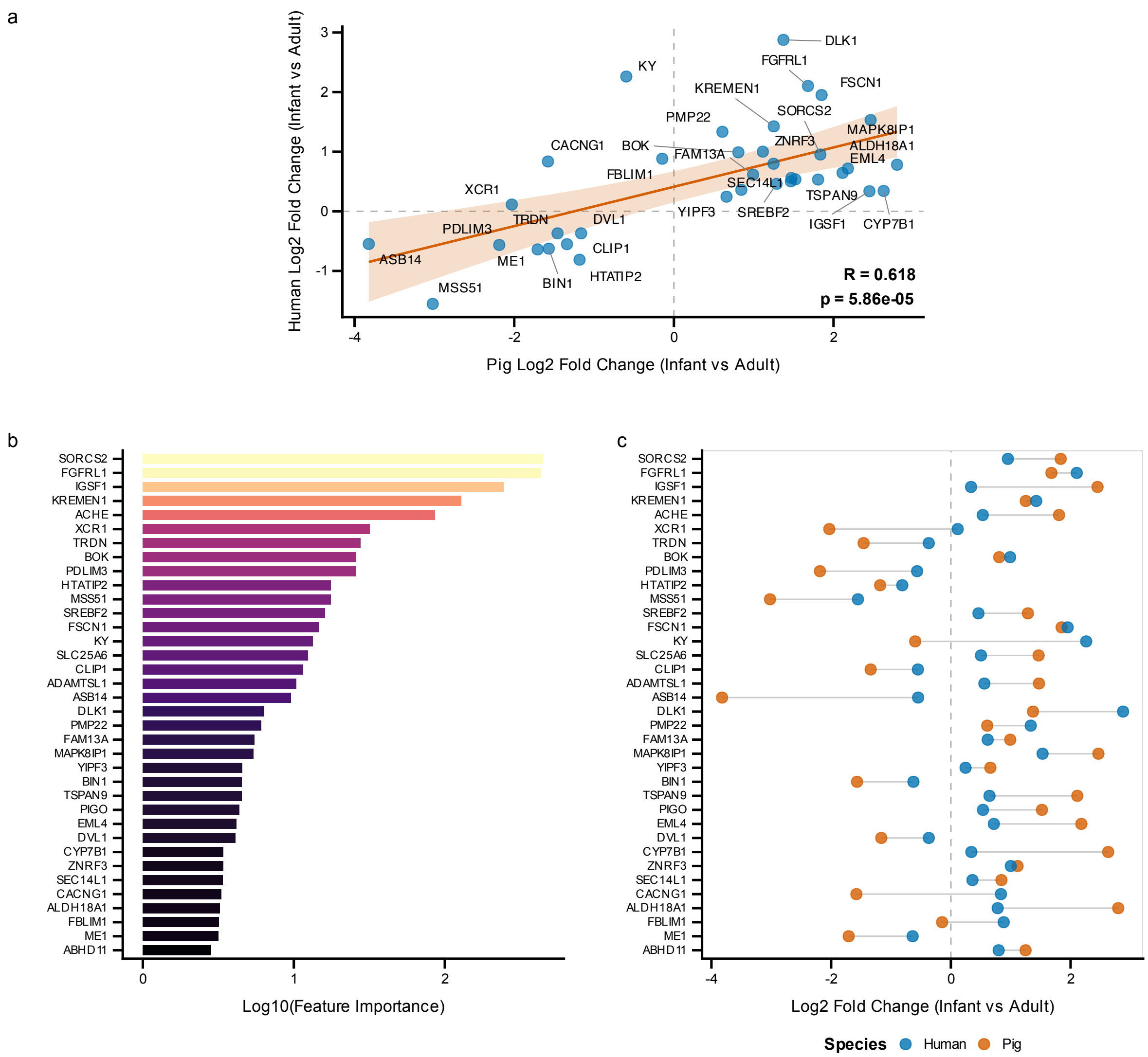

Figure 4: **Cross-species validation of developmental gene expression.** Human skeletal muscle transcriptomic data from Schaiter *et al.* [32] comparing infant (4–28 months) versus adult (19–65 years) samples were used to validate conservation of developmental signatures. (a) Scatter plot of $\log_2$ fold changes between pig and human, showing significant positive correlation (Pearson $R = 0.618$, $p = 5.86 \times 10^{-5}$, 95% CI: [0.364, 0.787]). 36 genes met significance criteria ($p < 0.10$) in both species, with 32 (88.9%) showing conserved direction of change. (b) Feature importance of top conserved markers in the developmental stage classifier. (c) Paired fold change comparison showing pig and human values for each conserved gene. Top markers include *SORCS2*, *FGFRL1*, *IGSF1*, *KREMEN1*, and *ACHE*.